Week5 Homework

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1. Bloch Sphere

a. We have $|\psi\rangle = \cos\frac{\pi/3}{2}|0\rangle + \sin\frac{\pi/3}{2}|1\rangle$, so $\theta = \pi/3$, $\phi = 0$, $\vec{r} = (\frac{\sqrt{3}}{2}, 0, \frac{1}{2})$.

b. $H|\psi\rangle = \frac{\sqrt{6}+\sqrt{2}}{4}|0\rangle + \frac{\sqrt{6}-\sqrt{2}}{4}|1\rangle$, $T|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + e^{i\pi/4}\frac{1}{2}|1\rangle$.

2. Evolution

a. We have $\frac{\mathrm{d}}{\mathrm{d}t}\alpha(t) = \frac{\omega}{i}\alpha(t)$, $\frac{\mathrm{d}}{\mathrm{d}t}\beta(t) = -\frac{\omega}{i}\beta(t)$, so $|\psi(t)\rangle = \frac{e^{-i\omega t}|0\rangle + e^{i\omega t}|1\rangle}{\sqrt{2}}$.

b. We have $|\psi(t)\rangle = \frac{e^{-i\omega t}|0\rangle + e^{i\omega t}|1\rangle}{\sqrt{2}} = \frac{e^{-i\omega t}+e^{i\omega t}}{2}|+\rangle + \frac{e^{-i\omega t}-e^{i\omega t}}{2}|-\rangle$. So $P(+) = (\frac{e^{-i\omega t}+e^{i\omega t}}{2})^2 = \cos^2(\omega t)$, $P(-) = \sin^2(\omega t)$. When $t = 0$, we have $P(+) = 1$, $P(-) = 0$.

3. Entanglement

a. We have $|a_0b_0 + a_0b_1 + a_1b_0 - a_1b_1| = |a_0(b_0 + b_1) + a_1(b_0 - b_1)|$. Let $c_0 = b_0 + b_1$, $c_1 = b_0 - b_1$, we have $|a_0b_0 + a_0b_1 + a_1b_0 - a_1b_1| = |a_0c_0 + a_1c_1|$. If $b_0 = b_1$, then we have $|c_0| = 2$ and $c_1 = 0$; if $b_0 \neq b_1$, we have $|c_1| = 2$ and $c_0 = 0$. So we have $|a_0c_0 + a_1c_1| = |a_0c_0| + |a_1c_1| = 2$. So this formula will always equal to 2. So the maximum is 2.

b. $|S| = \frac{1}{n}|\sum a_0^i b_0^i + \sum a_0^i b_1^i + \sum a_1^i b_0^i - \sum a_1^i b_1^i| = \frac{1}{n}|2n| = 2$. So the maximum is 2.

c. We first calc measure matrix A_0, A_1, B_0, B_1, that is a 2×2 matrix s.t. $A_0 a_0^{(+1)} = +1a_0^{(+1)}$, $A_0 a_0^{(-1)} = -1a_0^{(-1)}$ and so on. This deduces $\langle\psi'|A_0|\psi'\rangle$ is the expectation when use basis A_0 to measure $|\psi'\rangle$.

Let a basis C s.t. $C^{(+1)} = \cos\theta|0\rangle + \sin\theta|1\rangle$, $C^{(-1)} = \sin\theta|0\rangle - \cos\theta|1\rangle$. Let measure matrix $C = \begin{pmatrix} w & x \\ y & z \end{pmatrix}$. Let $c = \cos\theta$, $s = \sin\theta$. We have $C = P\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}P^{-1}$ where $P = \begin{pmatrix} c & s \\ s & -c \end{pmatrix}$ changes basis C to standard basis. Easy to see $P^2 = I$. So $C = \begin{pmatrix} c^2 - s^2 & 2cs \\ 2cs & -c^2 + s^2 \end{pmatrix} = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$. Since $\theta = \pi/4$ and $\theta = \pi/8$ for A_1 and B_0, we have $A_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $B_0 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$.

Let basis D s.t. $D^{(+1)} = \cos\theta|0\rangle - \sin\theta|1\rangle$, $D^{(-1)} = \sin\theta|0\rangle + \cos\theta|1\rangle$. Then $P = \begin{pmatrix} c & s \\ -s & c \end{pmatrix}$, $PP^T = I$ and $D = \begin{pmatrix} c^2 - s^2 & -2cs \\ -2cs & -c^2 + s^2 \end{pmatrix} =$

$\begin{pmatrix} \cos 2\theta & -\sin 2\theta \\ -\sin 2\theta & -\cos 2\theta \end{pmatrix}$. Since $\theta = 0$ and $\theta = \pi/8$ for A_0 and B_1, we have $A_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and $B_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$.

So we have $A_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $A_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $B_0 = \frac{A_0+A_1}{\sqrt{2}}$, $B_1 = \frac{A_0-A_1}{\sqrt{2}}$.

We have $A_0 \otimes B_0$ is the measure matrix for the combination of Alice's basis and Bob's basis (notice that $\otimes$ means Kronecker product). So $E(A_0, B_0) = \langle\psi|(A_0 \otimes \frac{A_0+A_1}{\sqrt{2}})|\psi\rangle = \frac{1}{\sqrt{2}}\langle\psi|(A_0 \otimes A_0)|\psi\rangle + \frac{1}{\sqrt{2}}\langle\psi|(A_0 \otimes A_1)|\psi\rangle$. So we need to calc $\langle\psi|(A_i \otimes A_j)|\psi\rangle$.

Notice that $A_1|0\rangle = |1\rangle$, $A_1|1\rangle = |0\rangle$.

We have $(A_0 \otimes A_0)|00\rangle = |00\rangle$, $(A_0 \otimes A_0)|11\rangle = (-1)^2|11\rangle = |11\rangle$, so $(A_0 \otimes A_0)|\psi\rangle = |\psi\rangle$, thus $\langle\psi|(A_0 \otimes A_0)|\psi\rangle = \|\psi\| = 1$. We have $(A_0 \otimes A_1)|00\rangle = |01\rangle$ and $(A_0 \otimes A_1)|11\rangle = -|10\rangle$, so since the result and input is orthogonal, $\langle\psi|(A_0 \otimes A_1)|\psi\rangle = 0$. Likewise, we have $\langle\psi|(A_1 \otimes A_0)|\psi\rangle = 0$ and $(A_1 \otimes A_1)|\psi\rangle = |\psi\rangle$ thus $\langle\psi|(A_1 \otimes A_1)|\psi\rangle = 1$.

So $E(A_0, B_0) = \frac{1}{\sqrt{2}}$, $E(A_0, B_1) = \frac{1}{\sqrt{2}}$, $E(A_1, B_0) = \frac{1}{\sqrt{2}}$, $E(A_1, B_1) = -\frac{1}{\sqrt{2}}$.

d. We have $S = 2\sqrt{2}$, which is impossible if we assume predetermined.